

Anabel Yong

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Summary

My current research focuses on Bayesian and probabilistic machine learning, with an emphasis on calibrated uncertainty and tractable surrogate models. In particular:

- **Sequential decision-making under uncertainty** - Bayesian optimization and related decision-theoretic frameworks, with an emphasis on reliable belief updating and uncertainty-aware decision rules
- **Uncertainty Calibration** - designing algorithms with confidence constraints for safe molecular design decisions
- **AI4Science applications** - applying these methods to molecular design, focusing on interpretability and data-efficient discovery.

Publications

- **Yong, A., Tripp, A., Hosseini-Gerami, L., Paige, B. (2025).** Bayesian Optimization for Molecules should be Pareto-Aware. (Accepted as Poster, NeurIPS 2025 AI for Science Workshop.) Link: [arXiv](#)

Education

MSc Comp. Statistics & Machine Learning, UCL (Distinction (Dean's List)) Sept 2023 - Sept 2024

Thesis Topic: *"Multi-Objective Bayesian Optimization with Independent Tanimoto Kernel Gaussian Processes for Diverse Pareto Front Exploration"*

Modules: Probabilistic & Unsupervised Learning (Gatsby PhD module) (91%), Machine Learning Seminar(76%), Numerical Optimization (80%), Statistical Natural Language Processing

BSc Biochemistry (with Mathematics), University of Edinburgh

Oct 2019 - June 2023

Thesis Topic: *"Evaluation of different False discovery rate approaches in Large-scale Proteomics Data"*

Modules: Bioinformatics (72%), Numerical Methods for Differential Equations (85%), Linear Programming, Modelling & Solutions (71%), Mathematics for Natural Sciences (80%)

Research Experience

- **Research Intern, National University of Singapore (PI: Jonathan Scarlett)** May 2025 - Sept 2025
 - Researching Bayesian optimization algorithms with diversity and safety constraints, focusing on uncertainty-aware batch selection for molecular design.
 - Benchmarked GP-BO variants and diversity constraints on molecular MPO tasks; reported diversity/utility trade-offs with $\#Circles/R^2$ /Hypervolume Indicator.
- **Machine Learning Research Intern, IgnitaLabs.AI (PIs: Dr. Austin Tripp, Cambridge MLG Group, Layla Hosseini-Gerami & Brooks Paige, UCL)** June 2024 – Sept 2024
 - Implemented a multi-objective Bayesian optimization algorithm using Tanimoto/MinMax Kernels with exact Gaussian Processes for small molecule optimization.
 - Benchmarked fixed weight scalarization(EI) versus standard Expected Hypervolume Improvement(EHVI) for molecular design.
- **Research Intern, Kustatscher Lab, University of Edinburgh** Oct 2022 - July 2023
 - Benchmarked a data pre-processing pipeline for proteomics, pgFDR, to enhance protein detection accuracy.
 - Aided in database curation of human proteins and microproteins of ProteomeHD2
- **Research Intern, ZiHeng Yang Lab, UCL** June 2022 - Sept 2022
 - Implemented Bayesian MCMC algorithms to model ABO blood type frequencies, contributing to population genetics.

Skills

Programming: Python, R, MATLAB, git, Linux, Google Cloud Platform (GCP), slurm, L^AT_EX

Machine Learning: PyTorch, JAX, BoTorch, GPyTorch

Bioinformatics/Cheminformatics: RDKit, PyMOL, UniProt, MaxQuant

Scholarships & Awards

Yayasan Sarawak Tun Taib Postgraduate MSc Scholarship 2023/2024

Value: $\approx$ £50,000

EPSRC Undergraduate Research Award (University of Edinburgh) 2021

Value: £2,500

References (available upon request)

- **Brooks Paige, Associate Professor in Machine Learning, University College London:** b.paige@ucl.ac.uk
- **Jonathan Scarlett, Associate Professor in Dept. of Computer Science & Mathematics, National University of Singapore:** scarlett@comp.nus.edu.sg
- **Layla Hosseini-Gerami, Chief Data Science Officer, IgnotaLabs.ai:** layla.gerami@ignotalabs.ai
- **Austin Tripp, Senior Machine Learning Research Scientist, Valence Labs:** austin.james.tripp@gmail.com